TELEPHONE NETWORKS

TELEPHONE NETWORKS

Any telecommunication network may be viewed as consisting of the following major
systems:

1. Subscriber end instruments or equipments
2. Subscriber loop systems
3. Switching systems
4. Transmission s:cstems
5. Signalling systems

TELEPHONE NETWORKS

A telephone network is a telecommunications network used for telephone calls between two or more parties.

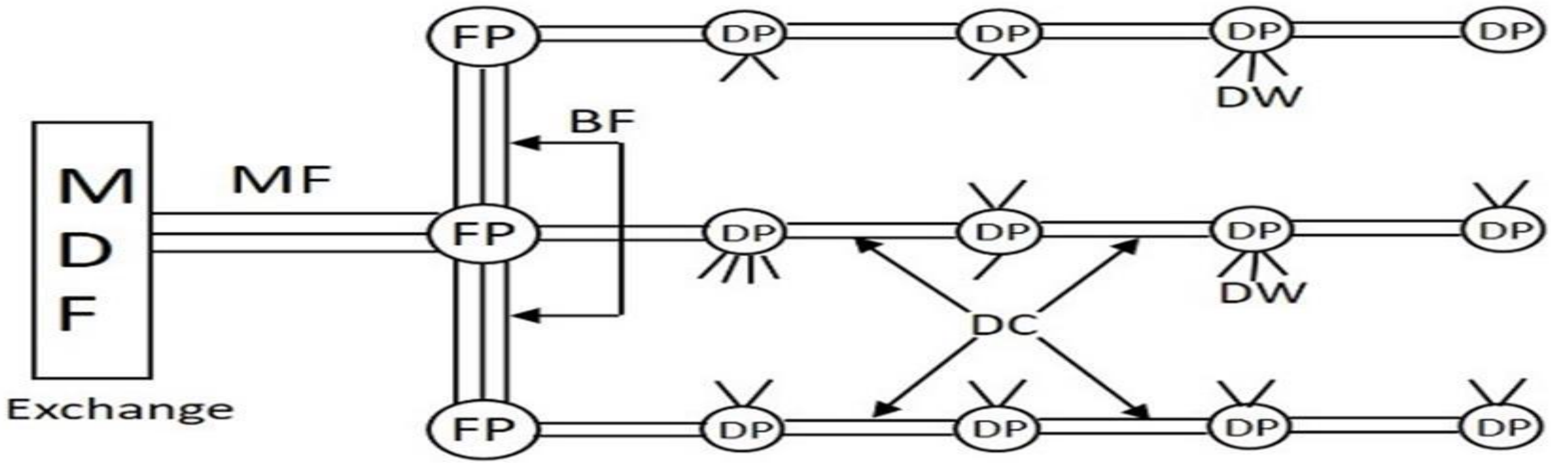
- There are a number of different types of telephone network:
  - ✓A landline network where the telephones must be directly wired into a single telephone exchange. This is known as the public switched telephone network or PSTN.
  - ✓A wireless network where the telephones are mobile and can move around anywhere within the coverage area.
  - ✓A private network where a closed group of telephones are connected primarily to each other and use a gateway to reach the outside world. This is usually used inside and call centers and is called a private branch exchange (PBX).

Subscriber Loop Systems

In a general telephone network, every subscriber has two dedicated lines connecting to the nearest switching exchange.

- which are called the **Loop lines** of that subscriber.
- The laying of lines to the subscriber premises from the exchange office is called **Cabling**

Subscriber loop systems



MDF = main distribution frame
DP = distribution point
DC = distribution cable
MF = main feeder
BF = branch feeder
FP = feeder point
DW = drop wires

Subscriber loop systems

- It is unwieldy to run physically independent pairs from every subscriber premises to the exchange.

- It is fast easier to lay cables containing a number of pairs of wires for different geographical location and run individual pairs as required by the subscriber premises.

- At the subscriber end, the drop wires are taken to a distribution point. The drop wires are the individual pairs that run into the subscriber premises on the MDF.

Switching Hierarchy and Routing

- The next important system in this is the switching hierarchy and routing of the telephone lines.
- The interconnectivity of calls between different areas having different exchanges is done with the help of **trunk lines** between the exchanges.
- The group of trunk lines that are used to interconnect different exchanges are called the **Trunk Groups**

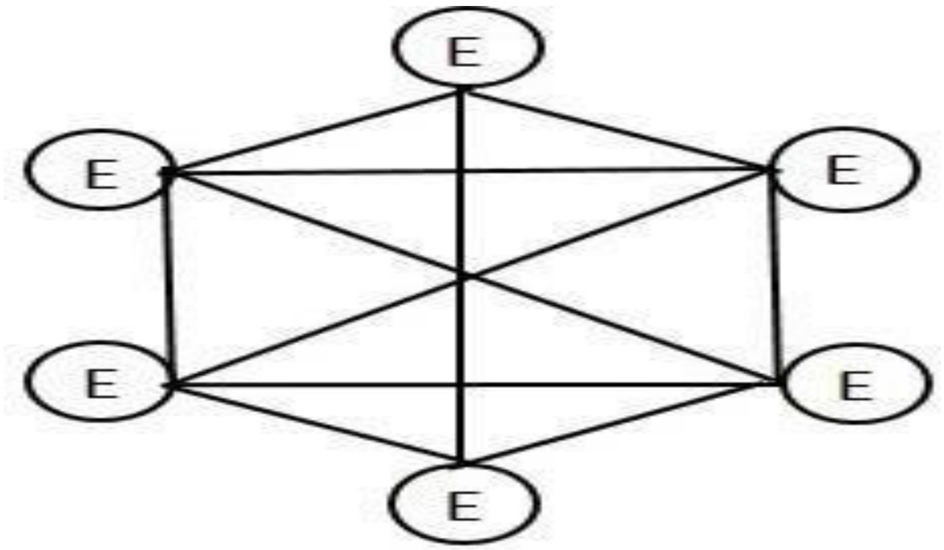
Switching Hierarchy and Routing

In the process of interconnecting exchanges, there are three basic topologies, such as

- Mesh Topology
- Star Topology
- Hierarchical

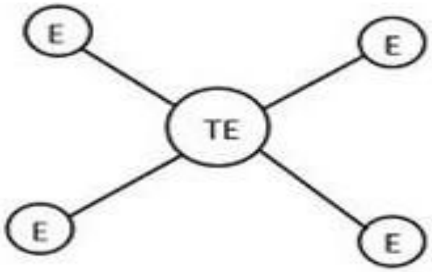
Switching Hierarchy and Routing

Mesh Topology

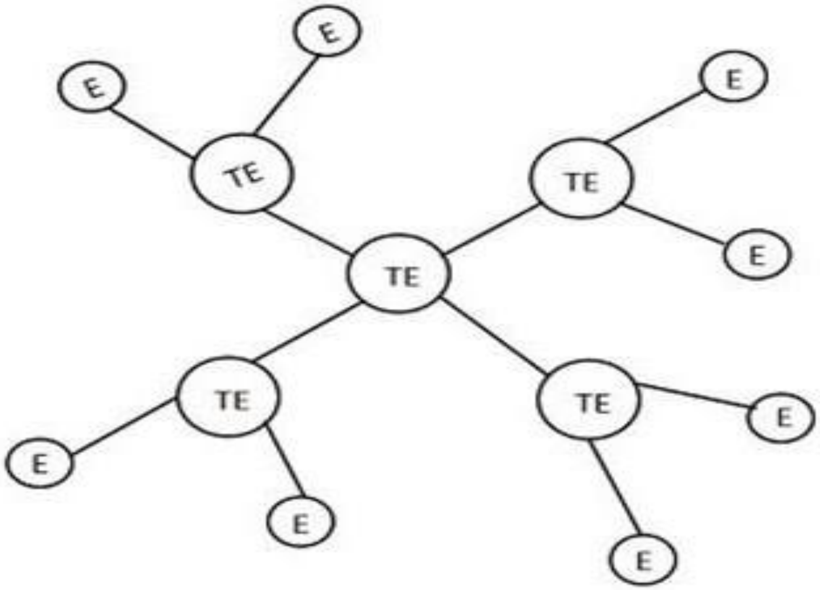


Switching Hierarchy and Routing

Star Topology



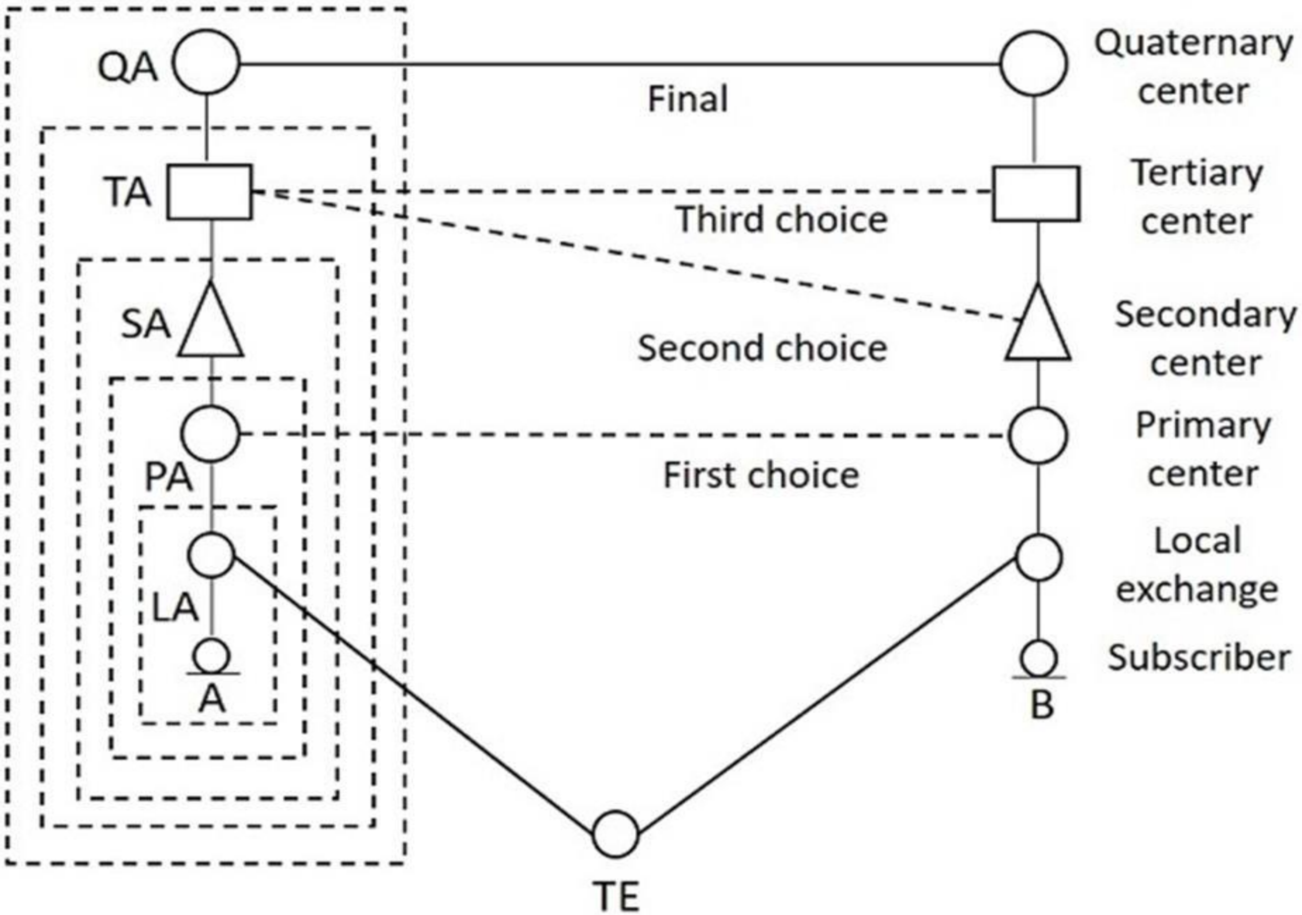
Star Topology



Two-level Star

Switching Hierarchy and Routing

Hierarchical



Transmission Systems

There are different types of transmission systems

- Radio systems
- Coaxial cable system
- Optical fiber system

Transmission Systems

- Fibre optic transmission systems have been discussed in detail
- we concentrate on radio and coaxial cable systems.
- Radio communication deals with electronic radiation of electromagnetic energy from one point to another through the atmosphere or free space.
- It is possible only in a certain portion of the electromagnetic frequency spectrum.

Transmission Systems

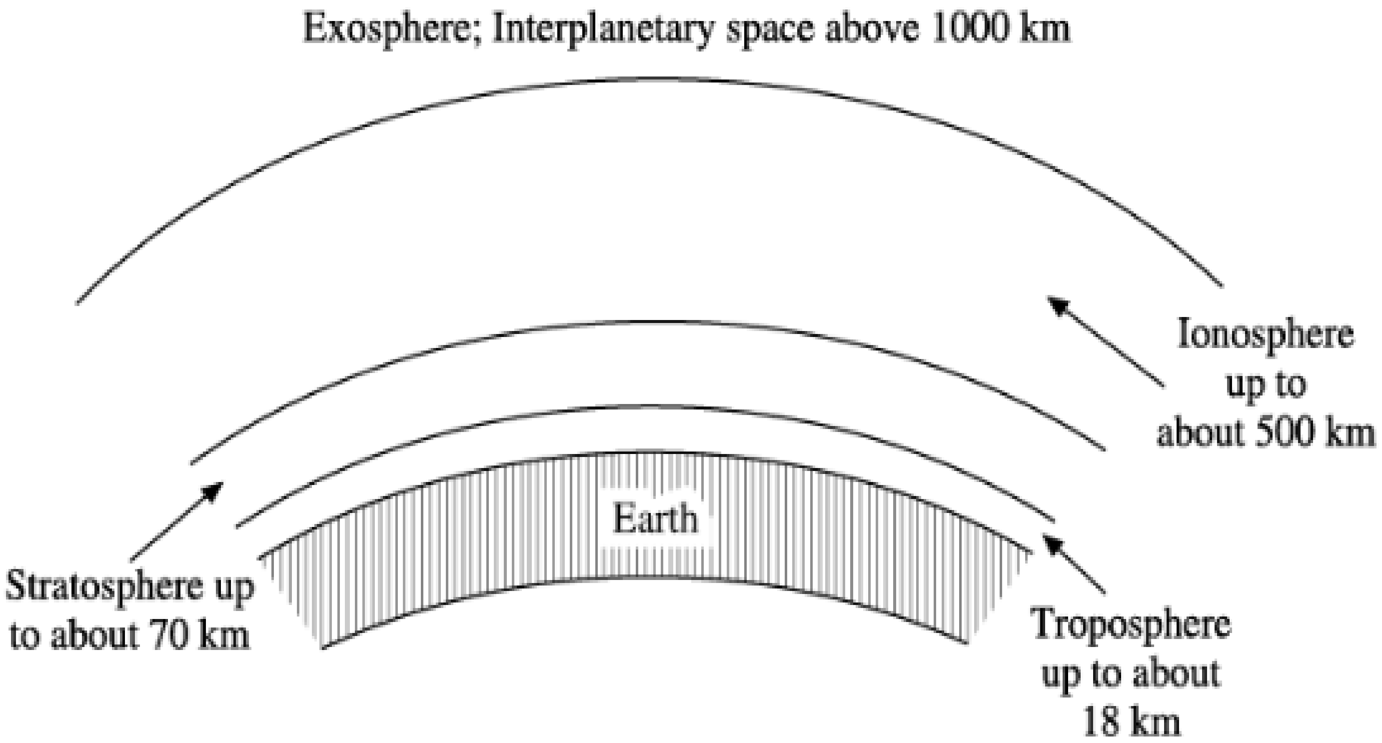
- This portion includes frequencies from 9 kHz to 4000Hz.
- While there are international allocations for the radio spectrum up to 275 0Hz, most of the commercial uses take place between 100 kHz and 20 0Hz.
- Different layers of the atmosphere play a role in propagating radio waves.

Transmission Systems

Depending upon the mechanism of signal propagation, the Radio communication has four varieties of communication, such as:

- Skywave or Ionospheric Communication
- Line-of-sight (LOS) microwave communication limited by horizon
- Tropospheric Scatter Communication
- Satellite Communication

Transmission Systems



Layers of atmosphere.

Numbering Plan

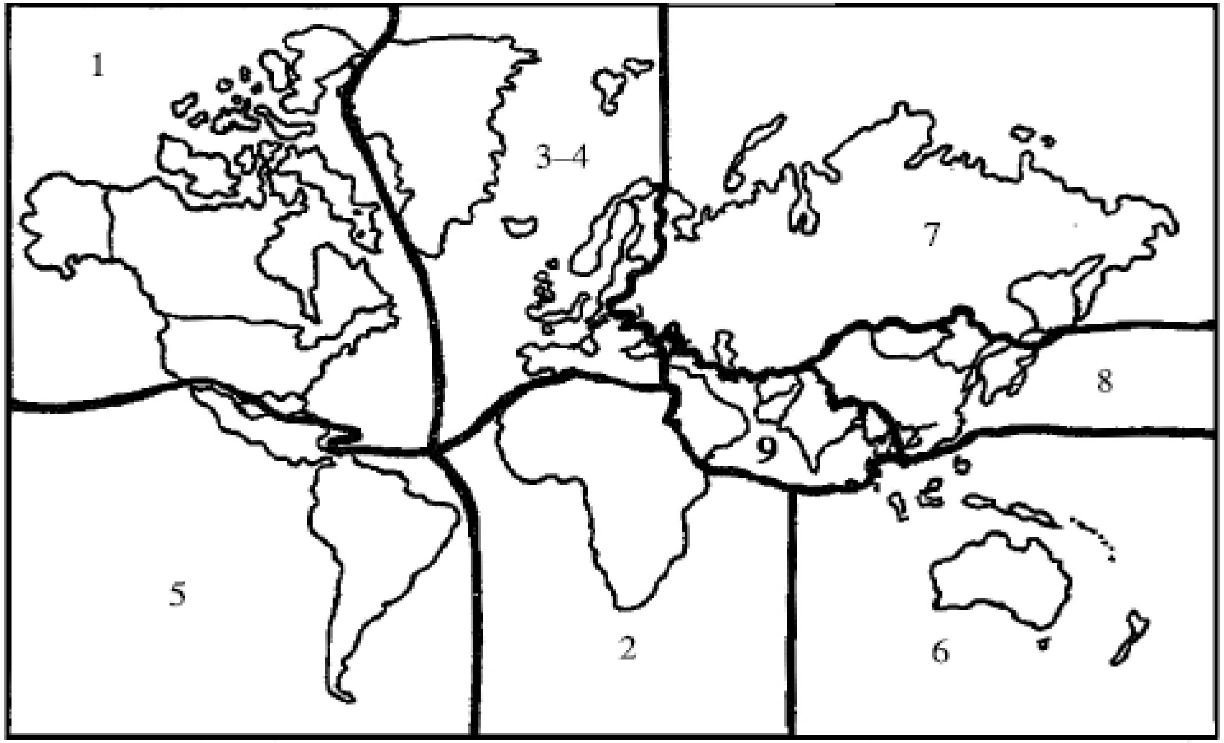
- Main Exchange: A large central exchange which serves the main business center of a town
- Satellite Exchanges: The smaller exchanges serving different localities .
- Multi-exchange area: The area containing the complete network of the main exchange and the satellites.

Numbering Plan

Types of Numbering Plans:

- **Open Numbering Plan**
- **Semi-Open Numbering Plan**
- **Closed Numbering Plan**

Numbering Plan

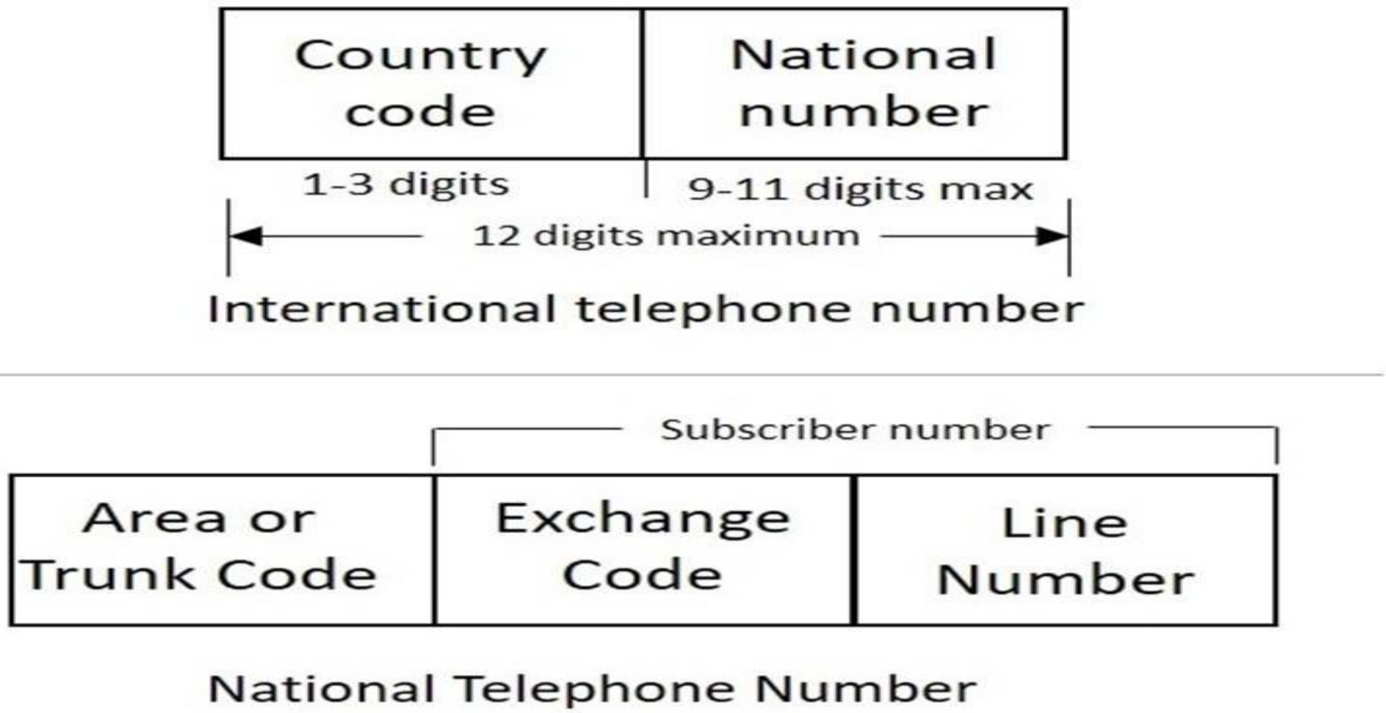


World numbering zones.

Numbering Plan

- Each zone is given a single digit code For the European zone, two codes have been allotted because of the large number of countries within this zone.
- Ever international telephone number consists of two parts as shown in Figure .
- The country code contains one, two or three digits, the first digit being the zone code in which the country lies.
- For example, in zone 3, France has the country code ‘33’ Albania ‘355’.
- In zone 9, India has the country code ‘91’ and Maldives 96O’.
- All the countries in the North American zone have the code as ‘1’

Numbering Plan



- ➢The Area Code or the Trunk Code
- ➢Exchange Code
- ➢Subscriber Line Number

Numbering Plan

Numbering Plan

- In the closed numbering scheme of North America, an area code consists of three digits, an exchange code three digits, and a subscriber line number four digits.
- Thus, a fixed 10-digit number forms the national number.
- In India where a semi open numbering scheme is used, a national number has 7—9 digits.
- But the apportionment between the STD code and the subscriber number varies widely.
- STD codes have 2—6 digits and the subscriber numbers 3—7 digits.
- The exchange codes are 1—3 digits long and the subscriber line numbers 2—4 digits.
- A numbering area in a region always has the region code as the first digit of the STD code.

  *STD: subscriber trunk dialling

Numbering Plan

For example, in region 2

- The city of Ahmedabad has the STDcode as ‘272’
- Nearby suburban town Bhopal has the code ‘2707’, and in regions, The city of Bangalore has the code ‘812’
- The hill town Tirumala the code 15747’.

Numbering Plan

There are four possible approaches to dialing procedures:

- Use a single uniform procedure for all calls. viz, local, national and international calls.
- Use two different procedures, one for international calls and the other for local and national calls.
- Use three different procedures, one for international calls, second for national trunk calls, and the third for local calls.
- Use four different procedures, three procedures same as given in 3 above and a fourth procedure for calls in the adjacent numbering areas.

Charging Plan

A charging plan for a telecommunication service levies three different charges on a subscriber:

1. An initial charge for providing a network connection
2. A rental or leasing charge
3. Charges for individual calls made

Charging Plan

The individual calls can be charged based on the following categories.

- Duration independent charging
- Duration dependent charging

Charging Plan

- In the olden days when STD and ISD facilities were not available, the trunk calls were established with the help of operators who were also responsible for the call charging .
- The subscriber meters are then useful only for local calls.
- To avoid the capital cost of providing meters and the operating costs of reading them at regular intervals and preparing the bills.
- some administrations have adopted a flat rate tariff system where some fixed charges for all estimated average number of local calls are included in the rental.
- This scheme is advantageous to subscribers who make a large number of calls but unfair to sparing users.

*ISD: International Subscriber Dialling

Signaling Techniques

There are three forms of signaling involved in a telecommunication network.

- Subscriber loop signaling
- Intra exchange or register signaling
- Interexchange or inter-register signaling

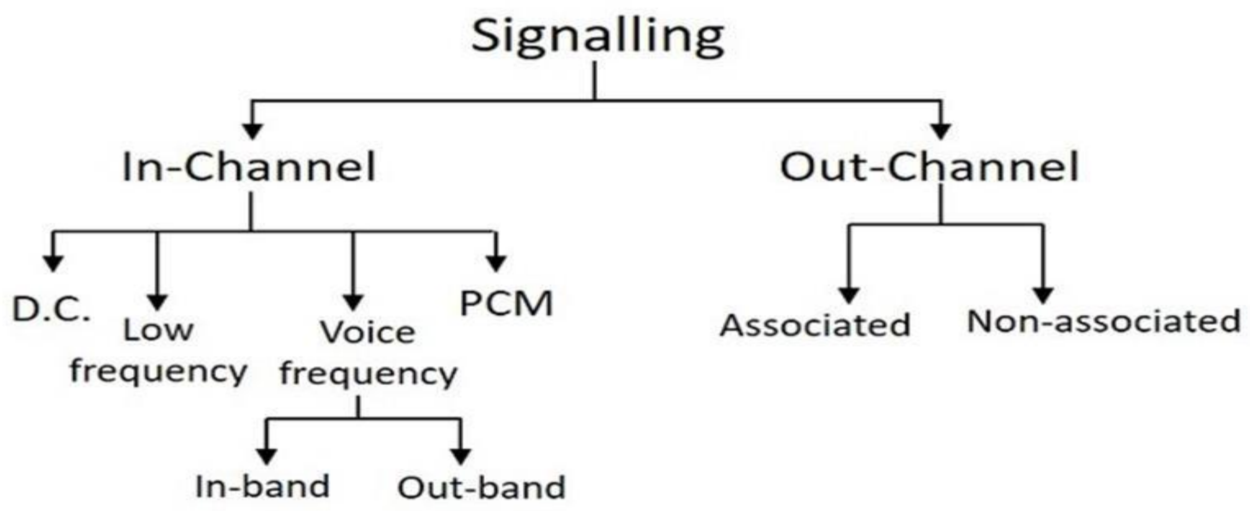
Signaling Techniques

There are three forms of signaling involved in a telecommunication network.

- Subscriber loop signaling
- Intra exchange or register signaling
- Interexchange or inter-register signaling

Line signaling: The network-wide signaling that involves end-to-end signaling between the originating exchange and the terminating exchange

Signaling Techniques



The two main types of signaling techniques are:

- In-Channel Signaling
- Common Channel Signaling

Signaling Techniques

In-Channel Signaling:

- It is also known as Per Trunk Signaling.
- This uses the same channel, which carries user voice or data to pass control signals related to that call or connection.
- No additional transmission facilities are needed, for In-channel signaling .

Signaling Techniques

Common Channel Signaling:

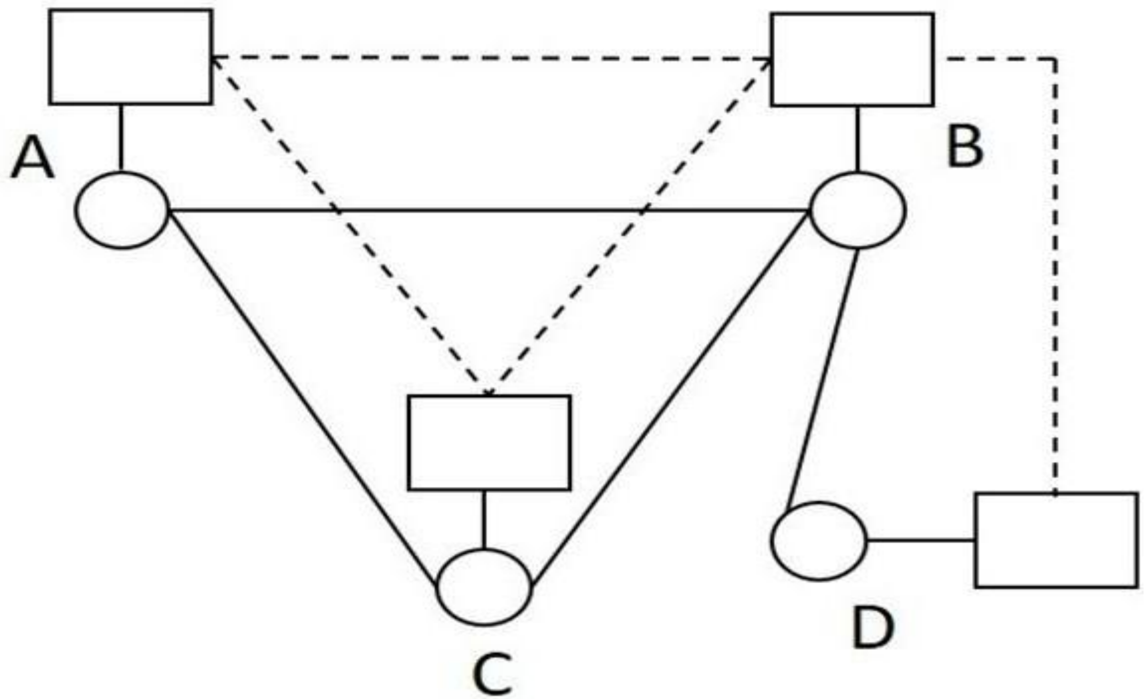
- It is also known as Per Trunk Signaling.
- This uses the same channel, which carries user voice or data to pass control signals related to that call or connection.
- No additional transmission facilities are needed, for In-channel signaling .

common channel signaling is implemented in two modes:

- Channel associated mode
- Channel non-associated mode

Signaling Techniques

Common Channel Signaling:

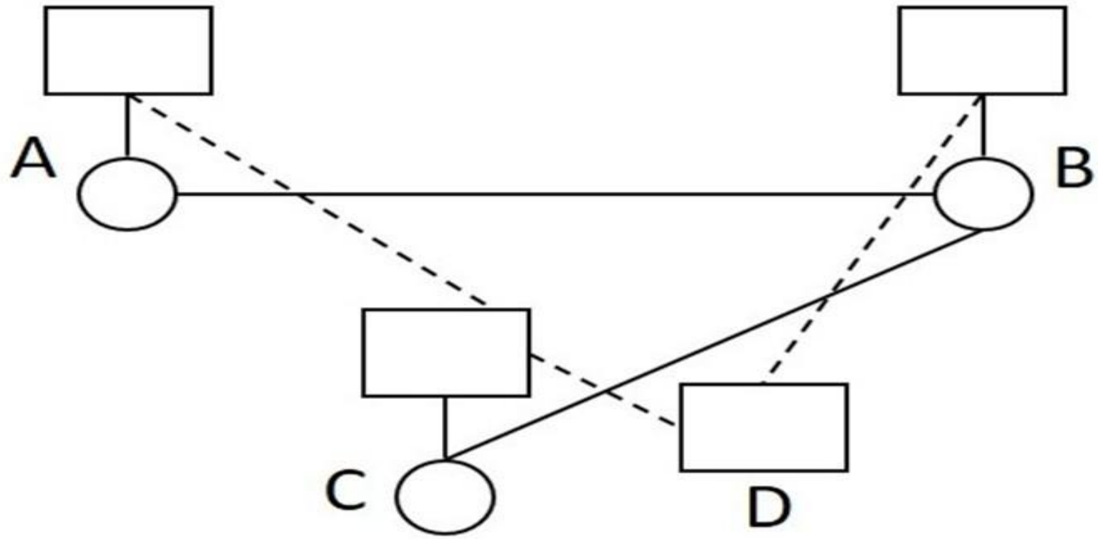


The signaling paths for the speech paths A-B, A-C-B and B-D are A-B, A-C-B and B-D respectively. The advantages of this signaling are:

- ➢The implementation is economic
- ➢The assignment of trunk groups is simple

Signaling Techniques

Channel Non-associated Mode



The signaling paths for the speech paths A-B and B-C are A-C-D-B and B-D-C respectively. The network topologies are different for signaling and speech networks

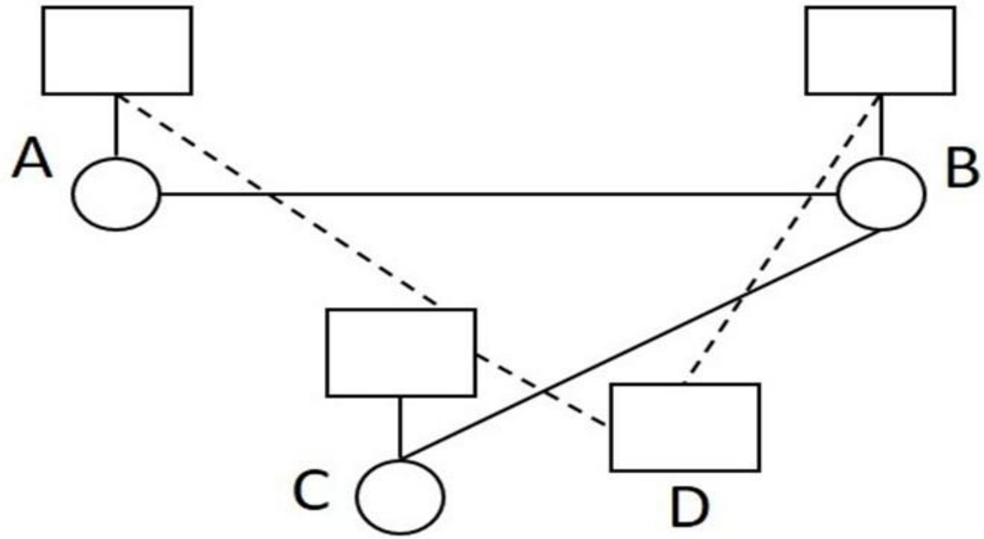
Signaling Techniques

In channel vs common channel

Inchannel	Common channel
Trunks are held up during signalling.	Trunks are not required for signalling.
Signal repertoire is limited.	Extensive signal repertoire is possible.
Interference between voice and control signals may occur.	No interference as the two channels are physically separate.
Separate signalling equipment is required for each trunk and hence is expensive.	Only one set of signalling equipments is required for a whole group of trunk circuits and therefore CCS is economical.
The voice channel being the control channel, there is a possibility of potential misuse by the customers.	Control channel is in general inaccessible to users.
Signalling is relatively slow.	Signalling is significantly fast.
Speech circuit reliability is assured.	There is no automatic test of the speech circuit.
It is difficult to change or add signals.	There is flexibility to change or add signals.
It is difficult to handle signalling during speech period.	There is freedom to handle signals during speech.
Reliability of the signalling path is not critical.	Reliability of the signalling path is critical.

Signaling Techniques

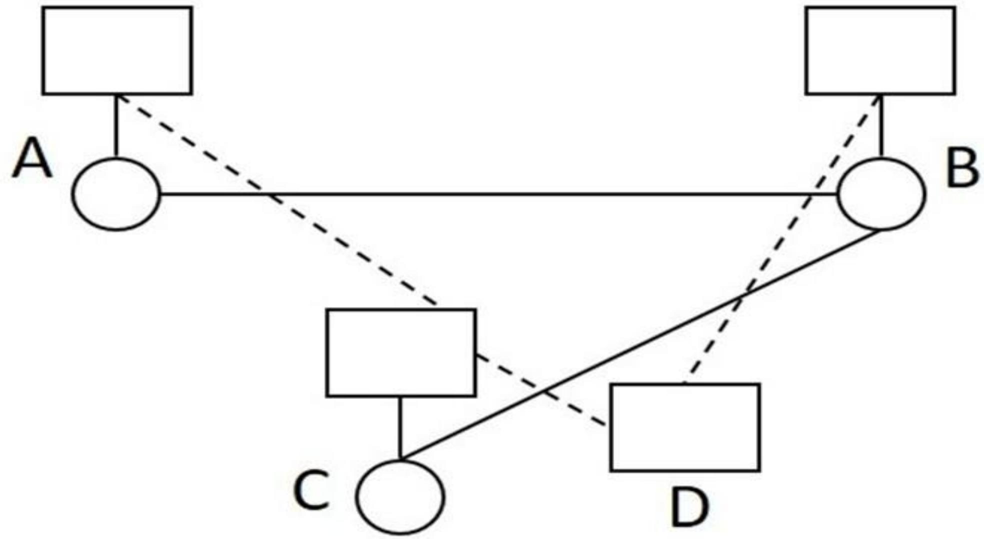
Channel Non-associated Mode



The signaling paths for the speech paths A-B and B-C are A-C-D-B and B-D-C respectively. The network topologies are different for signaling and speech networks

Signaling Techniques

Channel Non-associated Mode



The signaling paths for the speech paths A-B and B-C are A-C-D-B and B-D-C respectively. The network topologies are different for signaling and speech networks